

Mathematical Study of Radar Scattering from Heterogeneous Rough Soil Using the Finite-Element Method

Theoretical Foundations

Week 1

Outline

- 1 Physical Problem
- 2 Geometry of Rough Soil
- 3 Electromagnetic Formulation
- 4 Finite-Element Method
- 5 Boundary Conditions
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Motivation

- Radar backscatter is used to remotely sense **soil moisture** m_v .
- The electromagnetic response of soil depends strongly on its **dielectric properties**, which in turn depend on moisture content.
- Goal: understand and model how a radar wave scatters from a soil surface that is
 - *rough* (surface height varies), and
 - *heterogeneous* (moisture, hence permittivity, varies with depth).
- Underlying mathematical chain:

$m_v \rightarrow \epsilon_r \rightarrow \text{Maxwell's equations} \rightarrow \text{wave equation} \rightarrow \text{FEM} \rightarrow \text{field}$

Radar Scattering

- Radar transmits an incident electromagnetic wave toward the soil surface.
- At the surface, energy is reflected, transmitted into the soil, and scattered in many directions.
- The portion scattered back toward the radar is the **backscatter**.

incident wave → interaction with soil → scattered field

- The scattering strength is quantified by the **radar cross section (RCS)**.

Soil Moisture and Permittivity

- Volumetric soil moisture:

$$m_v = \frac{\text{volume of water}}{\text{volume of soil}}$$

- Water content strongly affects the relative permittivity ε_r :

$$\varepsilon_r = \varepsilon_r(m_v)$$

- Empirical silt-loam dielectric model:

$$\varepsilon_r(m_v) = (2.201 + 26.406 m_v + 34.563 m_v^2) - (-0.141 + 11.820 m_v + 8.100 m_v^2)$$

- If moisture varies with depth, $m_v = m_v(z) \Rightarrow \varepsilon_r = \varepsilon_r(z)$: the PDE coefficients become position-dependent.

Homogeneous vs. Heterogeneous Soil

Homogeneous soil

$$m_v(z) = m_0 \quad \Rightarrow \quad \varepsilon_r(z) = \varepsilon_{r,0} \text{ (constant)}$$

Heterogeneous soil

$$m_v(z) = \begin{cases} \text{varying function of } z, & 0 < z < d, \\ m_0, & z \geq d, \end{cases}$$

where d is the depth over which the moisture profile changes.

Heterogeneity makes the governing PDE's coefficients non-constant in space.

Geometrical Representation of Rough Soil

- A flat interface: $z = 0$. A **rough** interface: $z = f(x)$.
- The 2-D representation solves the field in the (x, z) -plane while the surface is rough only along x — far cheaper than a full 3-D simulation.
- Statistical description of a random rough surface:
 - RMS height h — magnitude of height fluctuations,
 - Correlation length l — horizontal scale of variation,
 - RMS slope $s = \frac{\sqrt{2} h}{l}$ (Gaussian-correlated surface) — measure of steepness.
- Ensemble (Monte Carlo) realizations of random surfaces are used to obtain statistically meaningful scattering behavior.

Maxwell's Equations

Time-harmonic ($e^{j\omega t}$), source-free Maxwell's equations:

$$\nabla \times \mathbf{E} = -j\omega\mu\mathbf{H}, \quad \nabla \times \mathbf{H} = j\omega\varepsilon\mathbf{E},$$

$$\nabla \cdot (\varepsilon\mathbf{E}) = 0, \quad \nabla \cdot (\mu\mathbf{H}) = 0.$$

- $\mathbf{E}$: electric field, $\mathbf{H}$: magnetic field
- ε : permittivity, μ : permeability, ω : angular frequency

Objective: determine the field inside the computational domain given an incident radar wave.

Vector Wave Equation

Eliminating one field from Maxwell's equations yields a scalar/vector wave equation for a field U :

$$\nabla \times \left(\frac{1}{p} \nabla \times U \right) - k_0^2 q U = 0,$$

where k_0 is the free-space wavenumber.

TM polarization

$$U = H_y, \quad p = \varepsilon_r, \quad q = \mu_r$$

TE polarization

$$U = E_y, \quad p = \mu_r, \quad q = \varepsilon_r$$

Soil moisture enters the wave equation through $\varepsilon_r = \varepsilon_r(m_v(z))$.

Why a Numerical Method Is Necessary

Sources of mathematical complexity that prevent closed-form solutions:

- ① Rough surface geometry $z = f(x)$ can be arbitrarily complicated.
- ② Material coefficient ε_r may depend on position.
- ③ The governing equation is a PDE, not an algebraic equation.
- ④ The physical domain is effectively unbounded.
- ⑤ Electromagnetic interface conditions must hold at a rough boundary.

⇒ A numerical method — the **Finite-Element Method (FEM)** — is required.

FEM: Fundamental Idea

- Replace the continuous domain with a finite number of small **elements**, connected at **nodes**.
- For 2-D problems, **triangular elements** represent complex/rough geometries well.

$$\Omega = \bigcup_{e=1}^{N_e} \Omega_e$$

- Rather than solving for the field at every point, FEM approximates it using a finite number of unknown coefficients tied to basis functions:

complicated continuous domain $\longrightarrow$ many small elements

Finite-Dimensional Approximation

Let $U(\mathbf{r})$ be the exact field. The FEM approximation is

$$U^h(\mathbf{r}) = \sum_{i=1}^N u_i N_i(\mathbf{r}),$$

- $N_i(\mathbf{r})$: basis (shape) functions
- u_i : unknown coefficients
- N : total number of degrees of freedom

This reduces the problem:

infinite-dimensional PDE $\longrightarrow$ finite-dimensional linear algebra, $\mathbf{u} = [u_1 \dots u_N]^T$

Basis Functions: 1-D Illustration

For a 1-D linear element with endpoints x_1, x_2 :

$$u(x) = N_1(x) u_1 + N_2(x) u_2,$$

$$N_1(x) = \frac{x_2 - x}{x_2 - x_1}, \quad N_2(x) = \frac{x - x_1}{x_2 - x_1}.$$

These satisfy the interpolation property

$$N_1(x_1) = 1, \quad N_1(x_2) = 0, \quad N_2(x_1) = 0, \quad N_2(x_2) = 1,$$

so the nodal unknown equals the approximate solution at that node.

2-D vector case: first-order **Whitney (edge) elements** serve as basis functions on triangular meshes for the electromagnetic field.

Weak (Variational) Formulation

Abstract PDE: $\mathcal{L}(U) = 0$. An approximate field U^h leaves a residual $R = \mathcal{L}(U^h) \neq 0$ pointwise.

Instead of requiring $R = 0$ everywhere, require a **weighted residual** to vanish for a test function T :

$$\int_{\Omega} T R \, d\Omega = 0.$$

For the electromagnetic problem (after vector identities + Green's theorem):

$$\int_{\Omega} \left[(\nabla \times T) \cdot \left(\frac{1}{\rho} \nabla \times U \right) - k_0^2 q T \cdot U \right] dS = \text{boundary terms}.$$

This integral (weak) form is the starting point for discretization.

Galerkin Method

Substitute $U^h = \sum_i u_i N_i$ and choose test functions from the *same* finite-element space: $T = N_j, j = 1, \dots, N$.

$$\sum_{i=1}^N u_i \int_{\Omega} \left[(\nabla \times N_j) \cdot \left(\frac{1}{\rho} \nabla \times N_i \right) - k_0^2 q N_j \cdot N_i \right] dS = \text{boundary contribution}$$

This produces one algebraic equation per test function j , giving the matrix system

$$\boxed{A\mathbf{u} = \mathbf{b}}$$

— the central result of the FEM discretization.

Element Matrices

The integral is evaluated separately over each triangular element Ω_e :

$$A_{ij}^{(e)} = \int_{\Omega_e} \left[(\nabla \times N_j) \cdot \left(\frac{1}{\rho} \nabla \times N_i \right) - k_0^2 q N_j \cdot N_i \right] dS + B_{ij}^{(e)},$$

where $B_{ij}^{(e)}$ collects boundary-term contributions from that element.

PDE $\rightarrow$ weak form $\rightarrow$ element equations $\rightarrow$ element matrices

Each local matrix involves only a small number of nodes/edges.

Global Assembly

- Neighbouring elements share nodes/edges, so local contributions combine at shared degrees of freedom.
- Local element matrices are summed into one global matrix:

$$A = \sum_{e=1}^{N_e} A^{(e)} \quad (\text{in global indexing}), \quad A\mathbf{u} = \mathbf{b}.$$

- Conceptual distinction:

local element matrices $\longrightarrow$ one globally coupled system

Although each element captures only local interactions, the assembled system couples the entire domain.

Sparsity of the Global Matrix

- Basis functions are **local** — nonzero only on a few neighbouring elements.
- Degrees of freedom in distant regions do not interact directly $\Rightarrow$ most entries $A_{ij} = 0$.

$$A = \begin{pmatrix} * & * & 0 & 0 & \dots \\ * & * & * & 0 & \dots \\ 0 & * & * & * & \dots \\ 0 & 0 & * & * & \dots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

This sparsity is what makes solving very large electromagnetic meshes computationally feasible using sparse direct solvers.

Mesh Resolution Requirement

The mesh must resolve the electromagnetic wavelength λ in the medium:

$$l_{\max} < \frac{\lambda}{20}.$$

- If elements are too large relative to λ , the field varies significantly within one element $\Rightarrow$ a low-order approximation becomes inaccurate.
- Trade-off:

smaller elements $\Rightarrow$ better resolution, but more DOFs $\Rightarrow$ higher cost.

The Unbounded-Domain Problem

- The physical radar-scattering problem is effectively unbounded (semi-infinite soil, open air region).
- A computer can only represent a **finite** computational domain

$$\Omega = \Omega_{\text{air}} \cup \Omega_{\text{soil}}.$$

- A naive reflecting outer boundary would send spurious reflections back into the domain, contaminating the solution.
- Need a boundary condition that mimics an **outgoing-wave** condition.

Absorbing Boundary Conditions (ABC)

- First-order ABCs approximate a purely outgoing scattered field U_s at the artificial outer boundary Λ :

$$(\nabla \times U_s) \times \hat{n} = jk_0 \cos \theta (\hat{n} \times (\hat{n} \times U_s))$$

- Enters the weak form directly and modifies the assembled system:

$$A = A_{\text{volume}} + A_{\text{boundary}}$$

- First-order ABCs are not perfectly reflectionless — this is a recognized source of numerical error (higher-order ABCs / PMLs can reduce it).

Incident and Scattered Fields

Total field decomposed as

$$U = U_i + U_s,$$

- U_i : known incident field
- U_s : unknown scattered field
- In the air region, the ABC is applied to U_s (incident field is known and subtracted out).
- In the soil region, the total field itself is required to be purely outgoing at the lower mesh boundary (no decomposition possible there).
- A **tapered** incident field is used so illumination fades near the lateral edges of the finite domain, reducing edge effects.

Conceptual FEM Pipeline

$m_v(z) \rightarrow \varepsilon_r(z) \rightarrow$ rough geometry $z = f(x) \rightarrow$ mesh
 $\rightarrow$ basis functions $\rightarrow$ weak form $\rightarrow$ Galerkin testing
 $\rightarrow$ element matrices $\rightarrow$ global assembly $\rightarrow A\mathbf{u} = \mathbf{b}$
 $\rightarrow U^h \rightarrow$ scattered far field $\rightarrow$ RCS

FEM is not a separate physical model — it is a **numerical framework** for solving the governing electromagnetic boundary-value problem.

Why FEM Is Appropriate Here

- Rough boundary $z = f(x)$ is naturally represented by an unstructured triangular mesh.
- Spatially varying permittivity $\epsilon_r(x, z)$ is incorporated directly into element integrals.
- No requirement for a simple analytical domain geometry.
- Locality of basis functions $\Rightarrow$ sparse global matrix $\Rightarrow$ large problems remain tractable.

Validation Strategy (Conceptual)

Numerical solutions should be checked against independent methods:

- **Small Perturbation Method (SPM):** independent reference for relatively smooth surfaces, valid roughly for

$$kh < 0.3, \quad s < 17^\circ.$$

- **Transfer Matrix Method:** independent reference for flat soil with depth-dependent moisture, approximating the heterogeneous profile as many thin homogeneous layers.

General principle:

FEM $\longleftrightarrow$ independent mathematical model, before trusting FEM on harder

Key Theoretical Takeaways

- 1 Soil moisture enters the problem only through the permittivity, $\varepsilon_r(m_v)$.
- 2 Governing physics reduces to a vector wave equation derived from Maxwell's equations.
- 3 FEM converts the continuous PDE into a finite linear system $A\mathbf{u} = \mathbf{b}$ via:
weak form $\rightarrow$ Galerkin testing $\rightarrow$ element matrices $\rightarrow$ assembly.
- 4 Locality of basis functions gives a sparse, tractable global system.
- 5 Absorbing boundary conditions truncate an unbounded physical domain.